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SUPERSEDING
MIL-W-52604(ME)
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MILITARY SPECIFICATION

WATER PURIFICATION UNITS, AND

WATER PURIFICATION EQUIPMENT SETS: 1,500 GPH;

PACKAGING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for materials, methods, and procedures for the preservation, packing, and marking of 1,500 gph van body mounted water purification units; 1,500 gph van body mounted water purification units mounted on a truck chassis; and 1,500 gph van body and chassis mounted water purification equipment sets.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-S-781	- Strapping, Steel, and Seals.
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
UU-S-48	- Sacks, Shipping, Paper.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

FSC 4610

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Military

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-B-131	- Barrier Material, Water Vaporproof, Flexible, Heat-Sealable.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-P-10603	- Pumps and Pumping Units, Centrifugal, Power-Driven, for Water, Packaging of.
MIL-T-14398	- Tanks, Fabric, Collapsible: Water, Nylon, Rubber-Coated.
MIL-E-16298	- Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-G-28554	- Generator Sets, Mobile Electric Power; Packaging of.

STANDARDS

Federal

FED. STD. No. 356	- Commercial Packaging of Supplies and Equipment.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer-Dollies; Preservation and Packaging of.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination and testing within the time frame specified (see 6.2) to prove, prior to starting production packaging, that the applied preservation, packing, and marking comply with the preparation for delivery requirements of this specification. Examination and tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3). The preproduction pack may be prepared utilizing either the preproduction model or a production water purification unit. When the preproduction model is utilized, any preservation and packing shall be removed by the supplier at no expense to the Government, when requested by the Government to facilitate comparison between the preproduction model and production water purification units.

3.2 Materials. Materials shall be as specified herein and in the applicable specifications. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.2.1 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116. CAUTION: Contact preservatives shall not be applied to any surface, other than the interior of the pump, which will come in contact with chemical solutions, slurries, or water, except waste water.

3.3 Preservation and unit packing. Preservation and unit packing shall be Level A, B, or Commercial as specified (see 6.2).

3.3.1 Level A.

3.3.1.1 Water purification unit, van body mounted, and van body and truck chassis mounted.

3.3.1.1.1 Actuating and hardware items. Surfaces of bolts, nuts, linkages, pins, grease fittings, levers, adjusting rods, clamps, capscrews, and couplings shall be coated with Type P-1 preservative, except when made of corrosion-resistant material. Surfaces of actuating and hardware items requiring a contact preservative not definitely specified herein shall also be coated with Type P-1 preservative.

3.3.1.1.2 Exposed ball bearings. Exposed ball bearings and adjacent surfaces of bearing mounts and shafts shall be coated with Type P-11 preservative. The coated parts shall be wrapped or covered with barrier material conforming to MIL-B-121, Type II, Grade A, Class 1, secured in place with tape conforming to PPP-T-60, Type IV.

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3.3.1.1.3 Exposed drive chains. Exposed drive chains shall be coated with enough Type P-3 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After excess preservative has drained, the entire chain shall be coated with Type P-1 preservative. To assure complete coverage, the chain shall be rotated during application of the preservative. Unprotected surfaces of the sprockets shall be coated with Type P-1 preservative.

3.3.1.1.4 Belts and pulleys. Belts shall be removed or released from tension. Unpainted pulleys (unless made of corrosion-resistant material) shall be coated with primer conforming to TT-P-664. Removed belts shall be secured to the one of the pulleys from which it was removed.

3.3.1.1.5 Electric motors. Electric motors shall be preserved in accordance with the Level A requirements of MIL-E-16298 by the alternate method as specified therein for assembled machines, except for the solution feeder assembly motor (see 3.3.1.1.6).

3.3.1.1.6 Solution feeder assembly. The solution feeder assembly gear-reduction mechanism and motor housing shall be filled to the operating level with Type I, Grade 10 preservative lubricating oil conforming to MIL-L-21260. The gears shall be actuated to assure coating of all parts and surfaces within the housing with the oil.

3.3.1.1.7 Cable assemblies. The cable assemblies shall be individually coiled in a minimum coil not less than 10 times the cable diameter. The coils shall be secured with equally spaced wraps of cotton tape or twine.

3.3.1.1.8 Control cabinet. The door of the control cabinet shall be secured with 3/4 inch by 0.020 inch strapping conforming to QQ-S-781, Class 1, Type I or IV, Finish A. Nominal 1-inch-thick wood blocking shall be placed under the strapping to provide for proper tensioning, while protecting the cabinet.

3.3.1.1.9 Gate valves and drain cocks. Gate valves and drain cocks shall be left open.

3.3.1.1.10 Filter elements. Each filter element shall be unit packed in a close-fitting box conforming to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 125, minimum, style optional.

3.3.1.1.11 Sleeves. A flat separator pad of fiberboard or similar material shall be inserted inside each sleeve. The pad shall be not less than 1/2 inch thick by 4-3/8 inches wide and 22 inches long. Each sleeve shall be wrapped in Kraft paper and inserted into a heat-sealable bag fabricated from material conforming to MIL-B-131, Class 2 and heat sealed.

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3.3.1.1.12 Pumps. Pumps shall be preserved in accordance with MIL-P-10603, Level A.

3.3.1.1.13 Float. The float shall be packed in a close-fitting box conforming to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 125, minimum, style optional.

3.3.1.1.14 Accessories. Blind flanges, pipe caps, and capscrews shall be preserved and packed in accordance with MIL-P-116, Method IC-1. Hose clamps shall be secured in quantities of 6 with cotton tape or wire ties. Preservation for the paddle shall not be required.

3.3.1.1.15 Repair parts. The preservative application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve repair parts.

3.3.1.1.16 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, Method IC-1.

3.3.1.1.17 Maintenance and operational tools. Maintenance and operational tools shall be preserved and packed in accordance with PPP-P-40, Level A.

3.3.1.1.18 Ground rods. The copper wire shall be removed from each ground rod and coiled. The coil shall be tied with cotton tape, twine, or wire and secured to the ground rod.

3.3.1.1.19 Intermediate and consolidated unit packing. The cable assemblies, coiled and tied as specified, shall be placed in the erdlator. Filter elements, unit packed as specified herein, shall be intermediate packed in quantities of 6 in a close-fitting box conforming to PPP-B-636, V3c, style optional. The sleeves unit packed as specified herein shall be intermediate packed in quantities of 6 in close-fitting boxes conforming to PPP-B-636, W5c, minimum, style optional. Each box shall be waterproofed as specified for waterproofing of slotted style boxes in accordance with the appendix to the box specification. The paddle shall be consolidated packed in the accessory chest. The blind flanges, pipe caps, hexagon-head capscrews, and hose clamps shall be consolidated packed in the chemical solution tanks. Technical publications and maintenance and operational tools preserved as specified herein shall be consolidated packed together in quantities not to exceed the weight limitation in boxes conforming to PPP-B-636, style optional. Repair parts, preserved and unit packed as specified, shall be consolidated packed in quantities not to exceed the weight limitation in boxes conforming to PPP-B-636, style optional. Cushioning and blocking to prevent free movement and damage to the contents or interior of the chemical solution tanks shall be provided for all consolidated packing.

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3.3.1.1.20 Van body. The components of each van body shall be preserved in accordance with MIL-STD-281, Level A, in addition, the heater exhaust outlet shall be covered with barrier material conforming to MIL-B-121, Type II, Grade A, Class 1 and secured with tape conforming to PPP-T-60, Type IV.

3.3.1.1.21 Truck chassis. The components of each truck excluding the engine and engine accessories shall be preserved in accordance with MIL-STD-281, Level A (mobile). Surfaces enclosed within the truck brake-drums shall be coated with primer as specified therein.

3.3.1.1.21.1 Truck chassis for driveaway delivery. When drivcaway method of shipment is specified, requirements of 3.3.1.1.22 shall be followed except that brakedrums shall not be coated with primer.

3.3.1.1.22 Engine and engine accessories. The engine and engine accessories for Level A (mobile) shipment shall be preserved in accordance with MIL-E-10062, Type I.

3.3.1.1.22.1 Engine and engine accessories for driveaway delivery. When driveaway method is specified, engine preservation shall not be required.

3.3.1.1.23 Vehicle lifting kit. Unprotected metal surfaces of the vehicle lifting kit components shall be coated with Type P-1 preservative and the components placed in the metal box provided.

3.3.1.2 Water purification equipment set, van body and truck mounted.

3.3.1.2.1 Water purification unit. The water purification unit shall be preserved and consolidated as specified in 3.3.1.1.

3.3.1.2.2 Van body, truck chassis, engine, and engine accessories. The van body, truck chassis, engine, and engine accessories shall be preserved as specified herein for the water purification van body mounted unit and the van body and truck mounted unit.

3.3.1.2.3 Adapters and bushings. Each adapter and bushing shall be wrapped with barrier material conforming to MIL-B-121, Type II, Grade A, Class 1. The barrier material shall be secured with tape conforming to PPP-T-76, or heat sealed.

3.3.1.2.4 Cable. The power cable shall be coiled in a minimum coil, not less than 10 times the cable diameter. The coil shall be secured with cotton tape or twine.

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3.3.1.2.5 Comparator. The components of the comparator shall be placed in the case provided. The case containing the components shall be packed in a close-fitting container conforming to PPP-B-636, W6c, style optional. The box shall be waterproof sealed with tape in accordance with the appendix to the box specification.

3.3.1.2.6 Pump covers. Each cover shall be preserved in accordance with MIL-P-116, Method IC-1.

3.3.1.2.7 Generator set. The generator set shall be preserved and packed in accordance with MIL-G-28554, Level A.

3.3.1.2.8 Graduate. The graduate shall be wrapped with barrier material conforming to MIL-B-121, Type II, Grade A, Class 1. The barrier material shall be secured with tape conforming to PPP-T-76.

3.3.1.2.9 Indicator solution. The bottle of indicator solution shall be packed in a close-fitting box conforming to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 125, style optional.

3.3.1.2.10 Hoses and hose assemblies. The ends of hoses and hose assemblies shall be covered with barrier material conforming to MIL-B-121, Type II, Grade A, Class 1. The barrier material shall be secured with tape conforming to PPP-T-76. The cotton hoses shall be coiled in a minimum coil and secured with a tie of cotton tape or twine.

3.3.1.2.11 Key set. The keys shall be placed in the case provided, and the case containing the keys shall be preserved in accordance with MIL-P-116, Method IC-1.

3.3.1.2.12 Extension lights. The lights shall be individually wrapped with barrier material conforming to MIL-B-121, Type II, Grade A, Class 1. The barrier material shall be secured with tape conforming to PPP-T-76.

3.3.1.2.13 Measures. The measures shall be individually wrapped with barrier material conforming to MIL-B-121, Type II, Grade A, Class 1. The barrier material shall be secured with tape conforming to PPP-T-76. Nesting of wrapped measures is desirable to conserve space.

3.3.1.2.14 Nozzles. The nozzles shall be individually wrapped with barrier material conforming to MIL-B-121, Type II, Grade A, Class 1. The barrier material shall be secured with tape conforming to PPP-T-76. When nozzles are furnished in a vendor's carton, no further packing shall be required.

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3.3.1.2.15 o-Tolidine dihydrochloride. The bottle of o-Tolidine shall be packed in a close-fitting box conforming to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 125, style optional.

3.3.1.2.16 Pumps. The pumps shall be preserved in accordance with MIL-P-10603, Level A.

3.3.1.2.17 Siamese connection. The Siamese connection shall be wrapped with barrier material conforming to MIL-B-121, Type II, Grade A, Class 1. The barrier material shall be secured with tape conforming to PPP-T-76.

3.3.1.2.18 Sodium arsenite. The bottle of sodium arsenite shall be packed in a close-fitting box conforming to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 125, style optional.

3.3.1.2.19 Suction strainer. The suction strainer shall be wrapped with barrier material conforming to MIL-B-121, Type II, Grade A, Class 1. The barrier material shall be secured with tape conforming to PPP-T-76.

3.3.1.2.20 Gate valves. Each gate valve shall be packed in a close-fitting box conforming to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 125, style optional.

3.3.1.2.21 Scoop. The scoop shall be wrapped with barrier material conforming to MIL-B-121, Type II, Grade A, Class 1. When the scoop is furnished in the vendor's carton, no further packing shall be required.

3.3.1.2.22 Wrenches. The open-end wrenches, pipe wrenches, and spanner wrenches shall be preserved in accordance with PPP-P-40, Level A.

3.3.1.2.23 Consolidated packing. The adapters, bushings, comparator, key set, Siamese connection, suction strainer, open-end wrenches, pipe wrenches, spanner wrenches, indicator solution, extension lights, o-Tolidine, sodium arsenite, and paddle furnished with the unit shall be intermediate packed in the accessory chest. The pails shall be nested, and the nozzles and scoop shall be placed in the nested pails. The cable assemblies furnished with the unit shall be removed from the erdlator and the rubber hose assemblies shall be placed in the erdlator. The cable assemblies and cotton hoses shall then be placed on top of the rubber hose assemblies in the erdlator. The gate valves, graduate, measures, clamps, blind flanges, bolts, pipe caps, and strainer bowl shall be consolidated packed in the chemical solution tanks. Cushioning and blocking to prevent free movement and damage to the contents or interior of the chemical solution tanks shall be provided for all consolidated packing.

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3.3.1.2.24 Cargo trailer. The trailer shall be preserved in accordance with the Level A (mobile) requirements of MIL-STD-281. The interior surface of the brakedrums shall be coated with primer as specified therein. When driveaway delivery is specified, brakedrums shall not be coated with primer.

3.3.1.2.25 Tanks. The tanks shall be preserved and packed in accordance with MIL-T-14398, Level A.

3.3.2 Level B.

3.3.2.1 Water purification unit, van body mounted. Each complete water purification unit shall be preserved, packed, and consolidated packed as specified for Level A with the following exceptions.

3.3.2.1.1 Exposed drive chains. Exposed drive chains shall be coated with enough Type P-3 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings.

3.3.2.1.2 Van body. The components of each van body shall be preserved and packed in accordance with Level B (mobile) requirements of MIL-STD-281. The heater exhaust outlet is to be covered and secured as specified in 3.3.1.1.21.

3.3.2.2 Water purification, van body mounted unit, truck chassis mounted.

3.3.2.2.1 Van body mounted unit. The van body mounted unit shall be preserved and packed as specified in 3.3.2.1.

3.3.2.2.2 Truck chassis. The components of each truck excluding the engine and engine accessories, shall be preserved in accordance with MIL-STD-281, Level B (mobile). Surfaces enclosed within the truck brakedrums shall be coated with primer as specified therein.

3.3.2.2.2.1 Truck chassis for driveaway delivery. When driveaway method of shipment is specified, requirements of 3.3.2.2.2 shall be followed except that brakedrums shall not be coated with primer.

3.3.2.2.3 Engine and engine accessories. The engine and engine accessories for Level A (mobile) shipment shall be preserved in accordance with MIL-E-10062, Type I.

3.3.2.2.3.1 Engine and engine accessories for driveaway delivery. When driveaway method is specified, engine preservation shall not be required.

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3.3.2.2.4 Vehicle lifting kit. Unprotected metal surfaces of the vehicle lifting kit components shall be coated with Type P-1 preservative and the components placed in the metal box provided.

3.3.2.3 Water purification equipment set, truck mounted. Each water purification equipment set shall be preserved and consolidated packed as specified for Level A, with the following exceptions.

3.3.2.3.1 Van body, truck chassis, engine, and engine accessories. The van body, truck chassis, engine, and engine accessories shall be preserved as specified herein for Level B, for the water purification van body mounted unit and the van body and truck mounted unit.

3.3.2.3.2 Cargo trailer. The components of each cargo trailer shall be preserved in accordance with MIL-STD-281, Level B (mobile).

3.3.2.3.3 Water purification unit. The water purification unit shall be preserved, packed, intermediate and consolidated packed in accordance with the Level B requirements specified herein.

3.3.2.3.4 Generator set. The generator set shall be preserved and packaged in accordance with MIL-G-28554, Level B.

3.3.3 Commercial.

3.3.3.1 Water purification unit, van body mounted; van body and truck mounted; and water purification equipment set; truck mounted. Preservation and unit packing of each water purification unit or water purification equipment set shall be in accordance with FED. STD. No. 356.

3.4 Packing. Packing shall be Level A or Commercial, as specified (see 6.2).

3.4.1 Level A.

3.4.1.1 Blocking, bracing, and cushioning. Blocking, bracing, and cushioning shall be provided as required to prevent or control movement of packed items and components within the van body. Blocking, bracing, and cushioning shall conform to MIL-STD-1186 or as specified herein.

3.4.1.2 Strapping. Where strapping of items or components located within the van body is specified, strapping material shall conform to QQ-S-781, Class 1, Type I or IV, Finish A, 3/4 inch by 0.020 or PPP-S-760,

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Type II, 3/4 inch by 0.028 except that for the securement of the launder ring, strapping conforming to PPP-S-760, Type II, of any width with a minimum breaking strength of 250 pounds may be used at option of the supplier. Strapping of items or components that are not within the van body shall be as specified herein. Where the requirements for strapping items or components not within the van body are not definitely specified, strapping shall conform to QQ-S-781, Class 1, Type I or IV, Finish B, 3/4 inch by 0.020 inch.

3.4.1.3 Water purification unit, van body mounted, and van body and truck mounted.

(a) Loosen the clamp holding the flexible coupling of the effluent launder ring. Place wood blocking, nominal 2 inches by 4 inches by approximately 18 inches long, between the launder ring and bridge rails at the two points where rails and ring cross. Raise the level of the effluent launder ring with the leveling screws until the blocking is held securely against the bridge rails.
(b) Place wood blocking, nominal 2 inches by 4 inches by approximately 18 inches long, under the launder ring; parallel and in line with the blocking between the ring and rails. Strap the launder ring to the bridge rails at four points of contact.
(c) Secure low-water-level float assembly with wood block so that the assembly does not rest on the sensitive switch. Pack cushioning material conforming to PPP-C-843, Type I, Class B around the float to prevent movement.
(d) Lever type handles of quick-opening gate valves shall be secured with tape conforming to PPP-T-60, Type IV.
(e) Secure the accessory chest in the bracket provided.
(f) The float and filter sleeves shall be placed in the erdlator with the cable assemblies. The ground rods with ground wires shall be taped to the framing of the slurry feeder with tape conforming to PPP-T-97 or PPP-T-60, Type IV. The filter elements, technical publications, tools, and repair parts for van body mounted and van body and truck mounted units shall be placed in the front area of the van body floor in front of the filter units and strapped to the clips provided.
(g) The doors of the van body shall be closed and secured to prevent pilferage.

3.4.1.3.1 Van body (truck mounted). Van bodies (truck mounted) preserved as specified for Level A shall not require packing.

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3.4.1.3.1.1 Vehicle lifting kit. The vehicle lifting kit, preserved as specified shall be secured to the truck chassis in accordance with Figure 1. When driveaway delivery has been selected, the vehicle lifting kit shall be located and secured in the cargo trailer.

3.4.1.3.2 Van body (unmounted). Van bodies (unmounted) shall be provided with a wood rubbing strip under each skid. Each rubbing strip shall be a nominal 2- by 6-inch member laid flat, equal in length to the wood filler of the van body skid, and beveled full depth at 45 degrees on each end. A wood member (the same length as the 2- by 6-inch member) 3 inches wide and equal in depth to the van body skid, shall be nailed to the top of each rubbing strip, flush with the outside edge. The rubbing strip shall be positioned with the nailed member against the bottom of the van body along the outside edge of the skids with the 2- by 6-inch members extending under the skids. The nailed member shall be cut or notched for skid-mounting angles and protruding objects on the bottom of the van body. The rubbing strips shall be drilled and attached with capscrews which pass upward through the rubbing strips and are secured to the regular mounting plates incorporated in the van body and the mounting angles bolted to the skids. In addition, 3/8-inch lag screws shall be applied upward through the rubbing strips and also at points equal distance between capscrews. Plain washers shall be placed under the heads of the cap and lag screws with both the washers and screwheads countersunk in the rubbing strips.

3.4.1.4 Water purification equipment set.

3.4.1.4.1 Accessory chest. The accessory chest containing the preserved and unit packed items shall be secured in the bracket provided.

3.4.1.4.2 Cable. The 50-foot cable, coiled and tied as specified, shall be placed inside the erdlator on top of the rubber hose assemblies.

3.4.1.4.3 Pumps. The pumps shall be secured in the brackets provided on the extreme left rear of the van body floor. NOTE: When the gasoline-engine-driven pump is furnished for MAP shipment, it shall be packed in accordance with the Level A requirements of MIL-P-10603 and shipped at the same time as the complete water purification set.

3.4.1.4.4 Charcoal. The drum of charcoal shall be placed on the floor of the van body and strapped to the filter unit.

3.4.1.4.5 Pump covers. The pump covers, preserved and unit packed as specified herein, shall be secured with cotton tape or twine to one of the pumps.

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3.4.1.4.6 Rubber pails. The nested pails containing the nozzles and scoop shall be placed and secured with wire on one of the pumps.

3.4.1.4.7 Filter elements, tools, etc. The filter elements, tools, repair parts, and technical publications shall be located in the front area of the van body floor, strapped to the clips provided.

3.4.1.4.8 Diatomaceous earth and limestone. The diatomaceous earth and limestone shall be packed in a cleated-plywood box conforming to PPP-B-601, Overseas Type, Style I, grade optional, except that plywood for the sides and ends shall be of 3/8 inch thickness plywood and the weight limitation shall be waived. Strapping shall be adjacent to the skids and over the intermediate cleat. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, Finish B, 3/4 inch by 0.023, minimum. A box of approximate inside dimensions of 40 by 46 by 28 inch height has been found to be adequate in previous procurements; however, since the bag sizes of the contents are subject to change, the box size will depend on the bag sizes furnished. The only limitations to the box size are that it must be of a size to be loaded, blocked, and braced in the cargo trailer that is a part of the set (see 6.6).

3.4.1.4.9 Tanks. The tanks for the set shall be packed in accordance with MIL-T-14398, Level A. For MAP requirements, only, two of the packed tanks shall be placed in the front area of the cargo trailer edgewise with the length of the box running across the trailer width. The remaining tank shall be placed in the rear area of the trailer edgewise with its length dimension running across the trailer width. The tanks shall be secured to the trailer with strapping. The strapping shall be attached to the cover tiedowns. Blocking and bracing shall be provided to prevent shifting of the load.

3.4.1.4.10 Water purification unit, van body mounted. The water purification unit, van body mounted shall be packed as specified herein for Level A for the units (see 3.4.1.3).

3.4.1.4.11 Generator set. The generator set preserved as specified for Level A, shall be packed in accordance with MIL-G-28554, Level A. For MAP shipments only the generator set shall be shipped separately but to accompany the other set components.

3.4.2 Commercial.

3.4.2.1 Water purification unit. Each complete water purification unit, components, repair parts, tools, and technical publications shall be packed

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in accordance with FED. STD. No. 356. The vehicle lifting kit shall be secured to the truck chassis in accordance with Figure 1 as specified for Level A. When driveaway delivery has been selected, the vehicle lifting kit shall be located and secured in the cargo trailer.

3.4.2.2 Water purification equipment sets. Each complete water purification equipment set shall be packed in accordance with FED. STD. No. 356.

3.5 Depreservation guide. A depreservation guide shall be prepared, placed in a waterproof envelope marked "Depreservation Guide", and secured in a conspicuous location on the water purification unit or water purification equipment set. Unless otherwise specified, DA Form 2258 "Depreservation Guide for Vehicles and Equipment" shall be used.

3.6 Marking.

3.6.1 Military. Marking for Levels A and B shall be in accordance with MIL-STD-129.

3.6.2 Commercial. Marking for Commercial shall be in accordance with FED. STD. No. 356.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The supplier is responsible for insuring that supplies and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction pack inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 Preproduction pack inspection.

4.3.1 Examination. The preproduction pack shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Unit of product. For the purpose of inspection, a completed pack of any one level prepared for shipment shall be considered a unit of product.

4.4.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined for the defects marked "X" for the applicable level in Table I of 4.5.1. Acceptable Quality Level (AQL) shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. Packs shall be examined for the defects listed in Table I.

Table I. Classification of Defects.

Defect		Level A	Level B
101.	Materials, methods, or containers not as specified for preservation, intermediate or consolidated packing, or packing. Each incorrect material, method, or container shall constitute one defect.	X	X
102.	Preservatives applied to surfaces which will come in contact with chemical solutions and water, except waste water.	X	X
103.	Exposed ball bearings not preserved as specified.	X	X
104.	Exposed drive chains not preserved as specified.	X	X
105.	Belts not removed or released from tension as specified.	X	X
106.	Solution feeder assembly not preserved and tagged as specified.	X	X
107.	Electric motors not preserved in accordance with the referenced document.	X	X
108.	Float switch rod and float not secured as specified.	X	X
109.	Control cabinet not strapped as specified.	X	X
110.	Gate valves and drain cocks not left open as specified.	X	X

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Table I. Classification of Defects (Cont'd)

	Defect	Level A	Level B
111.	Filter elements not preserved and intermediate packed together in groups of 6 as specified.	X	X
112.	Sleeves not unit packed and intermediate packed as specified.	X	X
113.	Pumps not preserved in accordance with the referenced document.	X	X
114.	Truck and van body components not preserved in accordance with referenced documents.	X	X
115.	Engine and engine accessories not preserved in accordance with referenced document.	X	X
116.	Surfaces enclosed within the truck brakedrum not coated with primer.	X	-
117.	Float not packaged as specified.	X	X
118.	Intermediate and consolidated packing not as specified.	X	X
119.	Trailer not preserved in accordance with referenced document.	X	X
120.	Trailer brakes not coated with primer.	X	-
121.	Pump covers not preserved as specified.	X	X
122.	Generator set not preserved in accordance with referenced documents.	X	X
123.	Ends of hoses and hose assemblies not covered as specified.	X	X
124.	Components not wrapped in barrier material and secured with tape as specified.	X	X
125.	Tanks not preserved and packed in accordance with referenced documents.	X	X
126.	Effluent launder not strapped to rails and leveling rods secured as specified.	X	X
127.	Ferric chloride not strapped as specified.	X	X
128.	Diatomaceous earth not in Level A bags in accordance with referenced document.	X	X
129.	Charcoal not located as specified.	X	X
130.	Actuating and hardware item not preserved as specified.	X	X
131.	Accessories, repair parts, and technical publications not preserved and packed as specified.	X	X
132.	Components not blocked, braced, anchored, or arranged within the van body as specified.	X	X
133.	Unmounted van bodies not prepared for shipment as specified.	X	X
134.	Depreservation guide not prepared and attached as specified.	X	X
135.	Marking illegible, incorrect, incomplete, or missing.	X	X
136.	Diatomaceous earth and limestone not packed as specified.	X	X

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5. PREPARATION FOR DELIVERY

(Not applicable to this specification).

6. NOTES

6.1 Intended use. This specification is intended to be used for direct reference in contracts or orders and Section 5 of commodity specifications for the preservation, packing, and marking of 1500 gph, van body or truck mounted water purification units and 1500 gph truck mounted water purification equipment sets (SC 4610-97-CL-E05).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction pack (see 3.1).
(c) Level of preservation and unit packing and level of packing required (see 3.3 and 3.4). Level B preservation and unit packing is intended to provide limited but economical protection and should be specified only when it is determined the water purification units or water purification equipment sets will be held in covered storage.
(d) When other than DA Form 2258 shall be used (see 3.5).

6.3 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, unit pack, pack, and mark the water purification units and water purification equipment sets in accordance with this specification.

6.4 DA Form 2258. The contracting officer should arrange to furnish DA Form 2258 when requested by the supplier.

6.5 Location of materiel. The location specified in this specification for items of the unit as well as the set govern the location. The Chest Loading Plan is designed for loads to be placed in the chest under normal use conditions and do not govern the location of supplies for shipping and storage purposes.

6.5.1 Change in unit or set components. When the size or configuration of an item is changed so that it no longer can be physically located as specified, the item shall be preserved and unit packed as specified, but located and secured elsewhere within the confines of the van. When additional but like items are added, they shall be preserved and unit packed as specified for similar items and located as above.

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6.6 Diatomaceous earth and limestone container. It is not possible to recommend any specific box size for the diatomaceous earth and limestone. While the size of the limestone bag remains quite constant the bag sizes for the diatomaceous earth varies considerably from one supplier to another. Eagle Pitcher's "Celaton(Diatomite)"; Great Lakes Carbon's "Diatomaceous Earth"; and Johns-Manville's "Celite 535" all vary considerably in bag length, width, and thickness. Therefore the box size and shape will depend on the source of the supplies to be packed.

6.7 Sources of material.

(a) A known source of readily available barrier material in stock rolls of 200 linear yards conforming to MIL-B-121, Type II, Grade A, Class 1 is Thilmany Pulp & Paper Co., Kaukauna, WI.
(b) Polypropylene strapping meeting requirements of 3.4.1.2 is available in a 750 foot dispensing pack with buckles from Interlake, Inc., 13501 Perry Ave., Riverdale, IL 60627.

Custodian:

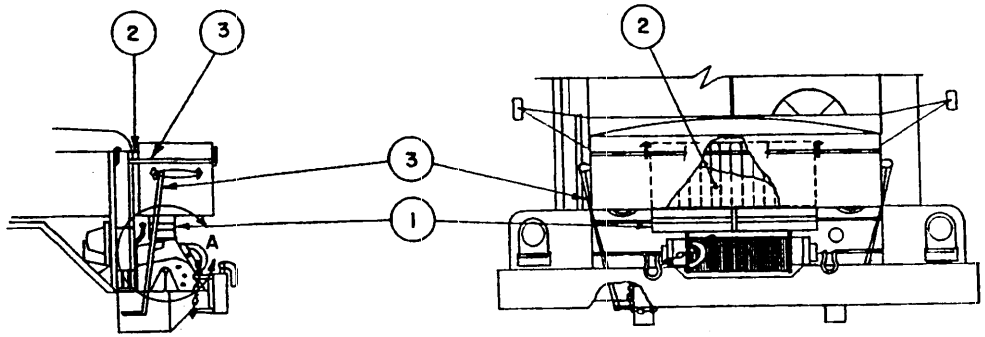
Army - ME

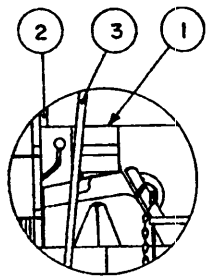
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DETAIL A

FIND NO.	QTY	NOMENCLATURE
1	AR	DUNNAGE LUMBER, 2" X 8" (NOMINAL).
		MINIMUM THICKNESS NECESSARY TO
		CLEAR WINCH OPERATING LEVER
		(APPROXIMATELY 3-1/4"). SET
		DUNNAGE INTO WINCH FRAME
		(SEE DETAIL A).
2	1	PLYWOOD DUNNAGE, 11" X 26" X 3/8"
		(MINIMUM). TO BE LOCATED BEHIND
		KIT CONTAINER AND IN FRONT OF
		RADIATOR GRILL. PLYWOOD TO
		CONFORM TO NN-P-530, GROUP A
		(PS 51, TYPE I, GRADE 3-4) OR
		GROUP B PS 1 STANDARD SHEATH-
		ING, WITH EXTERIOR GLUE).
3	3	STRAPPING AS SHOWN. STRAPPING
		TO CONFORM TO QQ-S-781,
		CLASS I, TYPE I OR IV, FINISH B
		3/4" X 0.020"

NOTE:
1. VEHICLE LIFTING KIT TO BE CENTERED ON WINCH IN FRONT OF RADITOR. DETAILS OF CONSTRUCTION OF KIT CONTAINER IN DRAWINGS D13217E6329 AND D13217E6330

Figure 1. LOCATION OF VEHICLE LIFTING KIT ON VAN MOUNTED UNITS.

DX1861A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R295

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-W-52604A(ME) Water Purification Units, and Water Purification Equipment Sets: 1,500 GPH; Packaging of

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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